CIR3: Comprehensive and Faithful Question-Answer Generation

Anonymous ACL submission

Abstract

Large Language Models (LLMs) excel at generating coherent and human-like questions and answers (QAs) across various topics, which can be utilised in various applications. However, their performance may be limited in domain-specific knowledge outside their training data, potentially resulting in low context recall or factual inconsistencies. This is particularly true in highly technical or specialised domains that require deep comprehension and reasoning beyond surface-level content. In this paper, we study the utility of LLMs in multi-agent settings to automatically generate high quality QAs from domain-specific documents. To this end, we propose, **C**ollective **I**ntentional **R**eading through **R**eflection and **R**efinement (**CIR3**), a QA-generator that leverages a collaborative and systematic process emphasising in-depth engagement with the input text. Our experiments indicate a substantial level of alignment between the CIR3-generated QAs and corresponding documents, while improving comprehensiveness by 21% and faithfulness by 17% compared to strong baseline approaches.

1 Introduction

Question-Answer Generation (QAG) is a data augmentation task that consists of generating a set of QA pairs given a context. QAG has a variety of applications, from information retrieval (Zhu et al., 2024) to healthcare (Shen et al., 2020), and education (Lindberg et al., 2013; Le et al., 2014; Kumar et al., 2015). Although Question Generation (QG) has been extensively researched in the context of language models (Uto et al., 2023; Meng et al., 2023), QAG presents a more challenging task, as it requires generating both the question and the answer, rather than assuming the answer is already provided in the input. While QG models offer a more direct and focused approach, they primarily focus on surface-level features of the context, such as facts and keywords. This is due to the limited amount of explicit information that is conditioned on the input answer. Furthermore, despite the proposal of various methods, generating comprehensive and semantically distinct questions from the same context remains under-explored as highlighted in (Zhang et al., 2022a; Vakulenko et al., 2023). The latter attributes this limitation to the lack of multi-reference training datasets that exhaustively cover all possible questions for each context. This inability is even more evident in highly technical or specialised domains, where documents are often rich in information.

In-Context Learning (ICL) (Brown et al., 2020) is an emerging paradigm that enables LLMs to learn new tasks without the need for extensive fine-tuning on specific data. By providing a description of the task, along with a few or even zero demonstrations as part of the input context, LLMs can be conditioned to perform well in various domains. This approach has shown promising results, surpassing state-of-the-art models in some tasks, and offers a potential solution to the challenge of limited data availability (Alaofi et al., 2023; Yuan et al., 2023). Despite the impressive results on popular NLP benchmarks, we find that using ICL for QAG, given a relatively complex document, often lacks robust inference mechanisms to deduce implicit relationships between the different key points inherent in the context. If the generation depends on comprehending the underlying connections that are not explicitly stated in the context, the model may fail to generate faithful QAs that accurately reflect this complexity. This is particularly problematic for information-dense contexts, which are common in highly domain-specific corpora, such as finance and science.

Recent advancements in LLM-based Multi-Agent (LLM-MA) systems have shown significant improvements in problem-solving abilities through planning, collaboration and autonomous task execution (Du et al., 2023; Liang et al., 2023). These

systems break down complex tasks into simpler sub-tasks to enhance complex task solving. Compared to standard LLMs and single-agent setups, LLM-MA systems offer advanced capabilities by leveraging collective intelligence and specialised skills (Guo et al., 2024). Motivated by the potential of these capabilities, we augment the QAG task with collective reasoning through the adoption of LLM-MA settings.

In order to address the aforementioned limitations in relation to generating comprehensive and faithful QAs from highly domain-specific documents, we derive a list of research questions around the adoption of LLM agents for QAG tasks:

R.1 : Can an LLM-MA workflow uncover deeper and perhaps implicit key concepts from a complex and information-dense document?

R.2 : How can LLM-MA effectively emphasise deep engagement, with a text, from different viewpoints to enable comprehensive and consistent generation and mitigate blind spots?

R.3 : (a) How can we incentive multiple agents to seek consensus? (b) how can we control the process of converge to reach common QAG, while avoiding premature collapse to uncomprehensive and/or unfaithful generation?

To address these research questions, we design CIR3 based on three corresponding hypotheses:

H.1 : Transactive reasoning enables the deduction of QAs that uncover the implicit relationships between the key concepts within the text.

H.2 : Multi-perspective group debate leads to an in-depth analysis of the document.

H.3 : Collective convergence, the process of a group of agents moving towards a shared output, requires disruptive signals to ensure diversity is maintained and collapse is avoided.

To build upon these hypotheses, CIR3 first utilises an optimised topology of information within the agents, as illustrated in Figure 1, to maximise the effectiveness of collaborative problem-solving and ensure an in-depth analysis of the input document. Second, CIR3 gains effectiveness by dynamically allocating specialised *writer* agents, each with a distinct perspective, based on the topic categories identified within the input context. Third, to reach a shared understanding of the document, despite the diverse perspectives and reasoning capabilities of the writers, CIR3 employs a *curmudgeon* agent as means to introducing variation. The curmudgeon, coupled with an external evaluation tool, incites the writers towards a balanced collective convergence on the key concepts within the text while maintaining diversity in the generated QAs.

While lexical matching is a standard evaluation method for QA tasks, its limitations become apparent when dealing with generative models, which often produce plausible answers not found in the predefined gold standard. This issue is further compounded by LLMs generating increasingly complex and lengthy answers, making lexical matching even less effective (Kamalloo et al., 2023). To ensure a comprehensive and accurate evaluation of CIR3, we employ diverse automatic metrics, in addition to human evaluation.

To summarise, our main contributions include: (1) We shift our focus from the typical task of QG to the more challenging QAG, which is inherently more difficult due to the limited search space and increased risk of producing duplicate QA pairs; (2) To the best of our knowledge, CIR3 is the first proposed QAG approach using multi-agent LLMs; (3) Our research demonstrates that incorporating an external signal significantly improves both the convergence rate and diversity maintenance within a group of agents. Specifically, in the context of QAG, this translates to optimising the number of iterations needed for the agents to reach collective agreement on the identified QAs pairs that comprehensively and faithfully cover the key concepts of the input text; (4) We demonstrate that CIR3 achieves improvements over strong baselines.

2 Related Work

QAG. Both rule-based (Heilman and Smith, 2010; Mostow and Chen, 2009; Huang and He, 2016) and neural (Alaofi et al., 2023; Du et al., 2017; Dhole et al., 2024) models have been extensively used for QG from text documents. Similarly, machine reading comprehension (Rajpurkar et al., 2016; Weissenborn et al., 2017; Min et al., 2018) has been employed for answer extraction (AE) from text given a question. However, traditional QG and AE methods produce either the question or the answer, unlike QAG which outputs both.

Several studies have leveraged pre-trained language models for QAG. These include fine-tuning BERT (Devlin et al., 2019) for AE and QG (Al-

berti et al., 2019), fine-tuning auto-regressive LMs for QG (Puri et al., 2020) using BART (Lewis et al., 2020) and RoBERTa (Liu et al., 2019) for AE, jointly fine-tuning LMs for AE and QG (Murakhovs'ka et al., 2022), and using QAG models to generate adversarial examples (Bartolo et al., 2021). Recent advancements have also focused on dynamically identifying question-worthy context words before using them to condition subsequent question generation (Vakulenko et al., 2023). (Ushio et al., 2023) improved QAG by designing three distinct approaches: Pipeline, Multitask, and End2end. (Zhang et al., 2022b) proposed combining entity linkage with a QA system, while (Puranik et al., 2023) enriched QA extraction by augmenting it with entity-level metadata.

Despite these advancements, current research predominantly focuses on specific question types, such as *Wh*-questions, rather than addressing open-ended questions. Furthermore, the focus tends to be on extracting short answers from existing text rather than generating comprehensive and detailed responses. Additionally, evaluating comprehensiveness of QAG remains an under-explored area. We tackle these challenges using CIR3.

LLM-based Multi-Agent Systems. Recent research has focused on LLM-based multi-agent systems to improve the quality of complex reasoning tasks. Studies like (Hong et al., 2023; Li et al., 2023; Qian et al., 2023) have shown that collaboration and task division among multiple agents can reduce hallucinations and generate more reliable outputs. Other works, such as (Chan et al., 2023; Shao et al., 2024), highlight the benefits of continuous debate among agents to correct misconceptions, analyse problems from diverse perspectives, and ultimately achieve higher-quality results.

Furthermore, prior research (Xiong et al., 2023) has examined the issue of inter-consistency through inter-agent negotiation. Similarly, drawing inspiration from robotics, (Chen et al., 2023) investigated consensus-seeking in multi-robot collaboration by analysing the effects of agent number, personality, and network topology. However, their work specifically focused on agent behaviour within a $1D$-space. Conversely, (Li et al., 2024) explored the concept of *flocking* where agents maintain proximity while avoiding collisions and preserving formations. In this work, we improve QAG task through a balanced collective convergence process.

3 Method

Given a context c consisting of a text passage, the task of QAG aims to produce a set of QA pairs, denoted as $\mathcal{G} = \{(q_i, a_i)\}_{i=1}^{N}$, that satisfies two crucial properties:

1. **Comprehensiveness:** The set $\mathcal{G}$ should cover all the key points and essential information present in the context c. In other words, for every significant aspect or piece of information $x \in c$, there exists at least one QA pair $(q_i, a_i) \in \mathcal{G}$ such that q_i elicits and a_i provides information relevant to x.

2. **Faithfulness:** Each answer a_i in $\mathcal{G}$ must be grounded in and supported by the factual content of the context c. This constraint ensures that the generated answers are not fabricated or hallucinatory, but rather reflect accurate information derived from the given text.

Formally, the QAG task can be formulated as an optimisation problem, where the objective is to find the set $\mathcal{G}$ that maximises both comprehensiveness and faithfulness with respect to the context c. This can be expressed as: $\mathcal{G}^* = \arg\max_{\mathcal{G}} [\text{Comp}(\mathcal{G}, c) + \text{Faith}(\mathcal{G}, c)]$, where $\text{Comp}(\mathcal{G}, c)$ and $\text{Faith}(\mathcal{G}, c)$ are scoring functions that assess the extent to which the set $\mathcal{G}$ covers the key points of c and adheres to the factual content of c, respectively.

In what follows, we describe CIR3 to generate the optimal solution $\mathcal{G}^*$ given c. This is achieved by building upon the aforementioned hypotheses to ensure that QAG is based on an in-depth analysis of the input text through an efficient flow of information and adoption of multiple views approach (3.1, 3.2), while maintaining QAG diversity and optimising the convergence rate of agents (3.3). The pseudo-code of the algorithm serving as the conceptual foundation of our approach is outlined in Appendix A.1: Algorithm Design.

3.1 Multi-Perspective Analysis

Incorporating multi-perspective or various viewpoints is crucial for analysing complex documents as it enhances the depth and breadth of understanding. Existing research highlights that a single perspective may introduce bias or overlook crucial aspects (Hall and Davis, 2007). For instance, STORM (Shao et al., 2024) emphasises the value of multiple perspectives in writing Wikipedia-like articles, by guiding participants to ask more in-depth questions in the pre-writing stage. Similarly,

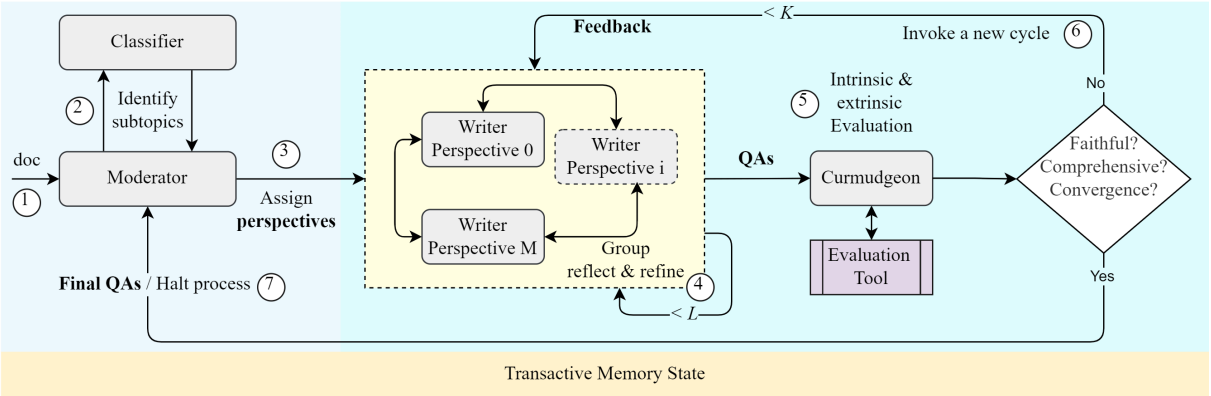


Figure 1: CIR3 takes an input document (1), identifies subtopics (2), and prompts writer agents to generate QA pairs based on their assigned perspectives (3). The QAs undergo iterative refinement by the writers (4), followed by an outer refinement where the curmudgeon, using its intrinsic knowledge and/or the evaluation tool, analyses the QAs and provides feedback for the next cycle (5, 6). The process halts when the curmudgeon is satisfied, and CIR3 returns the final QAs (7). The transactive memory serves as a central knowledge repository.

(Ihori et al., 2022) showcased how addressing various perspectives improved document clarity and readability in document revision task.

While STORM efficiently identifies different perspectives by surveying existing articles from similar topics using a search engine, CIR3 challenge is to discover diverse perspectives from a contained and limited context without retrieving external information. Given the input context c, CIR3 leverages LLM's language understanding capabilities to identify different subtopics within the input document c. To this end, we, first, utilise few-shot prompting, with a limited set of demonstrations, to guide a *classifier* agent to classify the context into M specific categories $P = \{p_1, ..., p_M\}$ (Figure 1 ①-②). For instance, given a finance-related document, CIR3 is prompted to discover the different $M = 3$ subtopics present in the context, such as *pensions*, *insurance*, and *savings*. Next, the *moderator* agent dynamically assigns each identified perspective p_j to a different writer W_{p_j}, while prompting the agents to analyse the input context and generate a set of QA pairs, $\mathcal{G}^{p_j}$, based on their respective perspectives (Figure 1 ③). Subsequently, as per 3.2 and 3.3, the list of $\mathcal{G}^{p_j}$ are aggregated into $\mathcal{Q}^+ = \{\mathcal{G}^{p_j}\}_{j=1}^{M}$, then subjected to iterative refinement and evaluation, ultimately resulting in $\mathcal{G}^*$. For better coverage of the overall information and the relationships between the key concepts within the context, CIR3 introduces W_{p_0} based on the corpus domain. Additionally, this approach guarantees at least one agent will be available even if no subtopics are identified.

3.2 Transactive Reasoning

(Momennejad, 2021) explores how the group structure, the pattern of connections between individuals, can significantly influence collective cognition and shared knowledge within the group. The author suggests that the structure of a network plays a crucial role in how memories are shared and aligned within a group. For instance, centralised networks, where information flows through a few key individuals, can lead to faster memory alignment but may also result in the loss of some details. In contrast, decentralised networks, with more diverse connections, may preserve a wider range of memories but take longer to reach consensus.

Drawing upon these insights, CIR3 employs a hybrid topology that consists of decentralised network of writer agents within a centralised network of two more agents, *moderator* and *curmudgeon*. To encourage communication and interaction, the group of writers form a fully-connected graph, where they operate at the same hierarchical level. To facilitate transactive reasoning, CIR3 adopts a reflection process, which benefits from the iterative exchange of critiques and refinements among the writers (Figure 1 ④). At iteration l, CIR3 gathers and aggregates feedback from all writers into $\mathcal{F}_l = \{\mathcal{F}_l{}^j\}_{j=1}^{M}$, links it to the previous QAs, $\mathcal{G}^+{}_{l-1}$, and then appends this updated information to the transactive memory. This creates a sequential memory state that evolves with each iteration: $\mathcal{M} = \{(\mathcal{G}^+{}_0, \mathcal{F}_1), ..., (\mathcal{G}^+{}_{l-1}, \mathcal{F}_l)\}_{l=0}^{L}$. The reflection prompt is specifically designed to encourage the participants to build upon each other's analysis, while maintaining comprehensive and faithful

output. To incentive the agents to seek an optimal consensus, CIR3 builds upon the group's decentralised graph to (1) capitalise on the strengths inherent in centralised networks, and (2) incite the group towards a shared and optimal solution (3.3).

3.3 Guiding Collective Cognitive Convergence

In addressing **R.3**, we take inspiration from the phenomenon of *Collective Cognitive Convergence (C3)* (Parunak et al., 2008, 2009) and from *How social network topology can shape collective cognition* (Momennejad, 2021) (details in Appendix A.2: Collective Cognition). C3 highlights that while convergence facilitates mutual understanding and coordination, if left unchecked, it can lead to cognitive collapse, by reducing the diversity of concepts to which the group is exposed, hence limiting the group's ability to explore other viewpoints and generate new ideas. The authors show that the lack of mechanisms to introduce variation incites the collapse and reduction in diversity.

In order to generate the optimal solution $\mathcal{G}^*$, CIR3 capitalises on:

(1) The strengths of combining decentralised and centralised networks, where (a) the information flow in the group of decentralised writers facilitates the preservation of a wider range of $\mathcal{G}^+$, which is amplified by the multi-perspective analysis, and (b) the rate of convergence in the broader centralised network (between (a) and the curmudgeon agent) facilitates a faster memory alignment of $\mathcal{G}^+$.

(2) The curmudgeon agent as means to introducing variation. Coupled with external evaluation tools (Figure 1 ⑤), the curmudgeon guides the writers towards a balanced collective convergence on the key concepts within the document, while maintaining diversity in the output.

Combined with the benefits of CIR3's hybrid topology, the cyclic process of reflection and refinement, between the writers and the curmudgeon (Figure 1 ⑥), amplifies the collective intelligence, and enables collaborative knowledge construction by sharing, discussing, and building upon each other's analysis, leading to a deeper understanding of the document. Additionally, this approach offers a solution to mitigate the disadvantages inherent in both centralised (potential loss of information) and decentralised (potential slow convergence) networks.

Once the inner-refinement cycle reaches either an agreement or the predefined maximum number of iterations, L, CIR3 is prompted to create a separate record of the latest refined QA pairs, $\mathcal{G}^- = \mathcal{G}^+{}_l$. This state is then passed to the outer-refinement cycle k, where the curmudgeon appends its feedback $\mathcal{F}\prime_k$ along $\mathcal{G}^-{}_{k-1}$ to the transactive memory, creating a central memory state that evolves with each outer-iteration of refinement: $\mathcal{H} = \{(\mathcal{G}^-{}_0, \mathcal{F}\prime_1), ..., (\mathcal{G}^-{}_{k-2}, \mathcal{F}\prime_{k-1}), (\mathcal{G}^*, \varnothing)\}_{k=1}^{K}$ where $\varnothing$ denotes a satisfactory alignment between the writers and the curmudgeon, which then routes the subsequent operation to the termination phase, through the moderator, yielding the final output $\mathcal{G}^*$ and halting the generation process (Figure 1 ⑦).

The curmudgeon is equipped with an evaluation tool to help quantifying the diversity of (a) generated questions, (b) generated answers, and (c) concatenated answers and input context. A lower diversity score in (c), combined with higher diversity scores in (a) and (b), would indicate high faithfulness and better coverage of the input context. To achieve this, we use *Vendi Score* (Friedman and Dieng, 2023) as an evaluation tool for diversity, where the objective is to minimise diversity in (c), while maximising it in (a) and (b).

At each iteration k, the curmudgeon initially analyses $\mathcal{G}^-$ to reason about the next action, whether to use the evaluation tool or to rely on its intrinsic knowledge to provide feedback. The result is then added to $\mathcal{H}$, which invokes another cycle of inner-refinements among the writers.

4 Experiments

4.1 Datasets

Existing QA datasets such as MS MARCO (Bajaj et al., 2018) and Natural Questions (Kwiatkowski et al., 2019) lack a comprehensive set of QA pairs for each context, making them unsuitable for our evaluation purposes. Therefore, our experiments focus solely on passages from two specific datasets from finance, FiQA (Maia et al., 2018) and InsuranceQA V2 (Feng et al., 2015), which we denote as InsurQA in this paper. For each dataset, a subset of 2000 passages is randomly chosen for our main experiments. Further details on the datasets can be found in Appendix B: Corpus Details.

4.2 Baselines

Prior research in this area has used varied experimental setups and has not focused on generating comprehensive sets of QA pairs from individual documents. As a result, direct comparisons be-

tween these works are challenging. Therefore, we establish the following baselines for our study:

1. **LLM-DP:** This baseline directly prompts `Meta-Llama-3-70B-Inst`[1] to generate QAs without explicit reasoning or tool utilisation. It serves as a measure of the LLM's ground performance.

2. **QGEN-AGEN** In this pipeline, we employ `query-gen-msmarco-t5-large-v1` from the Benchmarking IR BEIR (Thakur et al., 2021) to generate questions, which are then fed into `Meta-Llama-3-70B` to produce corresponding answers. This baseline assesses the LLM's performance when guided by an external query generation model.

4.3 CIR3 Implementation

We develop CIR3 using the LangGraph[2] library. In selecting models, we opt for the `Instruct` versions of `Meta-Llama-3-{70B,8B}` due to their outstanding performance. We set a temperature of 0.1 and a nucleus sampling of 0.5. To streamline the inference process, we utilise Groq API[3], which provides seamless integration. We also limit the number of generated QAs to 10 pairs per context, and the number of refinement iterations K, L to 6, 12, respectively. We release the code on GitHub[4], and include the illustrative prompts in Appendix E: Prompts.

In the current stage of our research, we focus on working with models from the same `Llama-3` family. However, we recognise the potential benefits of exploring a heterogeneous setup in future research, as this could provide valuable observations into the collaborative workflow in multi-agent settings.

4.4 Evaluation Metrics

Automatic evaluation of generated text remains a challenge as traditional metrics fail to align with human assessments. To address this limitation and provide a more comprehensive and refined evaluation of CIR3, we augment standard metrics with LLM-based scores tailored to our specific use case.

Statistical Scorers. We first use ROUGE-L (Lin, 2004), METEOR (Banerjee and Lavie, 2005), and *Jaccard Index* to calculate the scores between (1) the generated questions $\mathcal{Q}$ and the context c as reference, (2) the generated answers $\mathcal{A}$ and c, and (3) $\mathcal{Q}$ and $\mathcal{A}$. Then, we calculate the mean score over (1), (2) and (3), before calculating the average scores over each evaluation dataset.

Encoder-based Scorers. Beyond the token overlap, we use embedding-based similarity metrics, such as BERTScore (Zhang et al., 2020) and `BAAI/bge-large-en-v1.5`[5] (denoted with BGE score in this study). We measure the mean semantic scores between (1) c and $\mathcal{Q}$, (2) c and $\mathcal{A}$, and (3) $\mathcal{Q}$ and $\mathcal{A}$. To assess the quality of QAs when considered collectively, we also include BGE scores between (4) the concatenated questions $\mathcal{Q}_{\oplus} = \oplus_{i=1}^{N} q_i$ and c, (5) the concatenated answers $\mathcal{A}_{\oplus} = \oplus_{i=1}^{N} a_i$ and c, and (6) $\mathcal{Q}_{\oplus}$ and $\mathcal{A}_{\oplus}$.

LLM-based Scorers. To further quantify the comprehensiveness and faithfulness of the generated QA pairs, we adapt the G-EVAL (Liu et al., 2023) framework by merging the task definition and evaluation criteria prompt with a Chain-of-Thoughts (CoT) prompt (Wei et al., 2023) to specify detailed evaluation steps. This modification provides greater control over the assessment process compared to the original G-EVAL, where the LLM generates the CoT automatically. We evaluate the comprehensiveness of $\mathcal{G}^*$ based on *coverage*, *depth*, *accuracy* and *coherence*. Similarly, we evaluate the faithfulness based on *accuracy*, *exaggeration*, *consistency*, *justification*, *plausibility*, and *misrepresentation*. Additionally, we retain the G-EVAL scoring function, which normalises scores using a weighted sum of token probabilities in LLM output. We also used `GPT-4` with the temperature set to 0 to ensure reproducibility.

More details on the metrics used in this work can be found in Appendix C: Automatic Evaluation.

5 Results and Observations

5.1 Main Results

In all tables, the best-performing model is highlighted in **bold**, with the second-best underlined.

From Table 1, we observe that our approach outperforms both baselines on both datasets in terms of lexical metrics. Notably, CIR3 exhibits a relative improvement of 6.39%, 2.33%, 9.08% on METEOR, ROUGE-L and Jaccard Index, respectively, compared to the second best results. Although the observed overlap might not suggest a high degree

[1] https://ai.meta.com/blog/meta-llama-3
[2] https://langchain-ai.github.io/langgraph
[3] https://groq.com
[4] https://github.com/anonym-nlp-ai/cirrr
[5] https://github.com/FlagOpen/FlagEmbedding

Dataset	Model	METEOR				ROUGE-L (F1 Scores)				Jaccard Index			
		$s(c,\mathcal{Q})$	$s(c,\mathcal{A})$	$s(\mathcal{Q},\mathcal{A})$	Avg.	$s(c,\mathcal{Q})$	$s(c,\mathcal{A})$	$s(\mathcal{Q},\mathcal{A})$	Avg.	$s(c,\mathcal{Q})$	$s(c,\mathcal{A})$	$s(\mathcal{Q},\mathcal{A})$	Avg.
FiQA	LLM-DP	0.1571	0.3068	0.2119	0.2252	0.1951	0.3189	0.2781	0.2640	0.4377	0.5286	0.4881	0.4847
	QGen-aGen	0.1288	0.3383	0.1613	0.2094	0.1771	0.4003	0.2690	0.2821	0.4161	0.5391	0.4703	0.4751
	CIR3	0.1935	0.3791	0.2767	**0.2830**†	0.2153	0.3771	0.2893	**0.2939**	0.5511	0.6112	0.5983	**0.5868**†
InsurQA	LLM-DP	0.2422	0.3972	0.2717	0.3037	0.2877	0.4984	0.3447	0.3769	0.4784	0.5920	0.4987	0.5230
	QGen-aGen	0.1433	0.3134	0.1283	0.1949	0.1898	0.4903	0.2463	0.3088	0.3885	0.5749	0.4729	0.4787
	CIR3	0.3197	0.4391	0.3632	**0.3739**†	0.2950	0.4891	0.3972	**0.3937**	0.5261	0.6716	0.6104	**0.6027**†

Table 1: Evaluation results using standard metrics. † denotes significant differences ($p < 0.05$) from a paired t-test between **CIR3** and the best baseline LLM-DP.

Dataset	Model	BERTScore (F1 Scores)				BGE Semantic Similarity							
		$s(c,\mathcal{Q})$	$s(c,\mathcal{A})$	$s(\mathcal{Q},\mathcal{A})$	Avg.	$s(c,\mathcal{Q})$	$s(c,\mathcal{A})$	$s(\mathcal{Q},\mathcal{A})$	Avg.	$s(c,\mathcal{Q}_\oplus)$	$s(c,\mathcal{A}_\oplus)$	$s(\mathcal{Q}_\oplus,\mathcal{A}_\oplus)$	Avg.
FiQA	LLM-DP	0.8415	0.8597	0.8701	0.8571	0.6858	0.6847	0.7872	0.7192	0.7548	0.8078	0.8488	0.8038
	QGen-aGen	0.8339	0.8617	0.8472	0.8475	0.6932	0.7051	0.7358	0.7113	0.7462	0.8087	0.8183	0.7910
	CIR3	0.8702	0.9171	0.9088	**0.8987**†	0.8312	0.8542	0.8115	**0.8323**	0.8291	0.9118	0.9264	**0.8891**†
InsurQA	LLM-DP	0.8511	0.8810	0.8779	0.8700	0.7388	0.7540	0.8097	0.7675	0.8173	0.8948	0.8675	0.8598
	QGen-aGen	0.8282	0.8757	0.8472	0.8503	0.7231	0.7404	0.7344	0.7326	0.7708	0.8539	0.7487	0.7911
	CIR3	0.8972	0.9298	0.9175	**0.9148**†	0.7591	0.7736	0.8616	**0.7980**	0.8450	0.9395	0.9072	**0.8972**†

Table 2: Evaluation results using embedding-based metrics. † denotes significant differences ($p < 0.05$) from a paired t-test between **CIR3** and the best baseline LLM-DP.

of similarity, it is important to consider the limitations of lexical metrics, which are inherently less effective when evaluating generative tasks.

Further analysis in Table 2 shows that CIR3 consistently surpasses other models in semantic similarity metrics. CIR3 achieves an average improvement of 4.30% on BERTScore and 7.18% on BGE compared to the second-best model. This trend extends to contextual semantic similarity between the context and concatenated answers, suggesting that CIR3's generated answers are more faithful to the input text, potentially indicating lower hallucination and improved comprehensiveness.

These findings are further reinforced in Table 3 (LLM-based evaluation results) where CIR3 improves comprehensiveness and faithfulness with average scores of over 21% and 17%, respectively, compared to the second best model (LLM-DP).

Interestingly, LLM-DP demonstrates superior performance compared to QGen-aGen in all tests. This implies that the added query generator may not be beneficial, possibly due to the limitations of the T5 (Raffel et al., 2023) model in uncovering deeper key concepts in financial documents.

Dataset	Model	Comprehensive	Faithful	Avg.
FiQA	LLM-DP	0.7169	0.8030	0.7599
	QGen-aGen	0.5290	0.8414	0.6852
	CIR3	0.9312	0.9762	**0.9537**
InsurQA	LLM-DP	0.7317	0.8175	0.7746
	QGen-aGen	0.5560	0.8763	0.7161
	CIR3	0.9389	0.9879	**0.9634**

Table 3: LLM-based evaluation results for *comprehensiveness* and *faithfulness*.

	Aspect	LLM-DP	CIR3	p-value
Comprehensiveness	Coverage	0.7875	0.9375	0.0033
	Depth	0.7750	0.9125	0.0038
	Coherence	0.7625	0.9250	0.0023
	Avg.	0.7750	**0.9250**	
Faithfulness	Accuracy	0.7500	0.9125	0.0020
	Representation	0.7875	0.9125	0.0042
	Diversification	0.8250	0.8875	0.0104
	Avg.	0.7875	**0.9041**	

Table 4: Human evaluation results on 80 sets of QA pairs generated by CIR3 and LLM-DP. The ratings (1 to 5) are normalised between 0 and 1. The scores are analysed using a paired t-test (p-values are presented).

Our analysis also reveals, in Tables 1 and 2, that CIR3's questions are significantly more aligned with the context compared to both baselines. This indicates that the CIR3's deep engagement with the input document helps bridging the gaps in machine reading comprehension, which results in more comprehensive and relevant question generation.

The results presented in Tables 1, 2, 3, provide compelling evidence of the effectiveness of CIR3.

5.2 Human Evaluation

We further conduct human evaluation on 80 samples from the InsurQA corpus and the corresponding generated QA pairs by CIR3 and LLM-DP. We ask 8 experts in finance to assess 10 sets of QA pairs each, focusing on comprehensiveness and faithfulness. Comprehensiveness is evaluated based on three aspects: *coverage*, *depth*, and *coherence*. Similarly, faithfulness is assessed based on: *accuracy*, *representation*, and *diversification*. Each aspect is scored on a scale from 1 (worst) to 5 (best). Table 4 shows the average scores and paired

Model	Comprehensiveness	Faithfulness	Avg.	Context Distribution per Refinement Cycle Ranges (%)			
				$[0,3]$	$[4,6]$	$[7,9]$	$[10,12]$
LLM-DP	0.7399	0.8221	0.781	100	-	-	-
CIR3	**0.9451**	**0.9895**	**0.9673**	04.00	76.00	19.50	00.50
CIR3 w/o perspectives	0.9115	0.9653	0.9384	09.00	69.00	17.00	05.00
CIR3 w/o Curmudgeon	0.8370	0.9046	0.8708	17.50	57.00	14.00	11.50

Table 5: Effect of multi-perspective reasoning and Curmudgeon on *comprehensiveness* and *faithfulness*

t-test results, aligning with the findings in Table 3. CIR3 demonstrates significant improvement over the baseline LLM-DP, with an increase of 15% on comprehensiveness and 11.66% on faithfulness. Detailed information about the human evaluation process can be found in Appendix D: Human Eval.

5.3 Ablation Studies

To provide additional support for our hypotheses in **H.2** and **H.3**, we conduct an ablation study with two variations of CIR3: (1) **CIR3 w/o perspectives**. Following (Shao et al., 2024), in this variation, we aim to assess the impact of multi-perspective reasoning. We modify the moderator's prompt by removing the section that assigns diverse perspectives to the writer agents. To ensure a fair comparison, we maintain the same number of writers as in the original model (determined by the number of identified subtopics); (2) **CIR3 w/o Curmudgeon**. In this variation, we disable the curmudgeon agent to evaluate the effect of introducing external variation to the writer's sub-network.

For this study, we randomly select 200 samples, equally split between both datasets, and capped the refinement cycles between writers at 12 for each input. The results in Table 5 demonstrate that CIR3 surpasses the two alternative variations. Nonetheless, both variations outperform the baseline LLM-DP, providing some support for our hypotheses.

Effect of multi-perspective reasoning. Table 5 shows that *CIR3 w/o perspectives* yields inferior results compared to CIR3, suggesting that multi-perspective group debate contributes to a comprehensive and faithful output, as proposed in **H.2**.

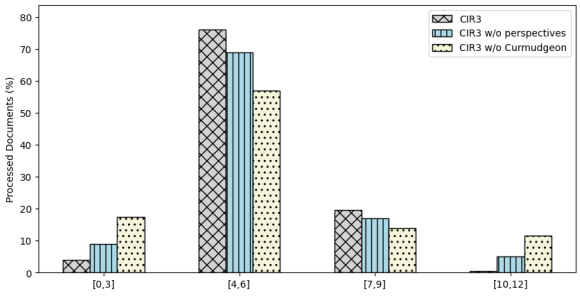


Figure 2: Number of inner-refinement cycles (x-axis), given as intervals, required to process the input documents (y-axis), given as percentage.

Effect of variation. Removing the disruptive signal, in *CIR3 w/o Curmudgeon*, significantly impairs performance, reducing faithfulness by 8.49% and comprehensiveness by 10.81%. This can be explained by examining the number of refinement cycles (given as intervals) required to process the input documents, as in Figure 2 and Table 5. Compared to CIR3, and *CIR3 w/o perspectives*, *CIR3 w/o Curmudgeon* shows a significant increase in the number of contexts falling within the refinement cycle ranges $[0,3]$ and $[10,12]$. For the interval $[0,3]$, *CIR3 w/o Curmudgeon* exhibits a 13.5% increase compared to CIR3 and an 8.5% increase compared to *CIR3 w/o perspectives*. Similarly, for the interval $[10,12]$, *CIR3 w/o Curmudgeon* shows an 11% increase over CIR3 and a 6.5% increase over *CIR3 w/o perspectives*. This aligns with **H.3**, where the absence of variation can result in either (1) a potential immature collective convergence (collapse) and loss of information, characterised by a small number of iterations and potentially low comprehensiveness scores, or (2) a potential slow convergence, characterised by a large number of iterations and a high likelihood of low truthfulness.

6 Conclusion and Future Work

This paper presents CIR3, a novel system designed to generate comprehensive and truthful set of QA pairs from information-dense documents. The core of CIR3 lies in its efficient flow of information, which employs LLM-based agents to conduct an in-depth analysis of the input context. This is achieved through a combination of transactive reasoning, multi-perspective assessment, and a balanced collective convergence process. Empirical evidence clearly demonstrates that CIR3 significantly outperforms baseline models in terms of performance. Future research will investigate the potential of a heterogeneous multi-agent framework and expand CIR3's applicability to various domains and tasks, including multi-modal applications.

Limitations

Our approach significantly outperforms baseline methods, but scaling CIR3 to handle a large number of writer agents poses challenges due to increased computational demands and the complexity of managing agent interactions. Furthermore, CIR3, which has its roots in multi-agent systems, can be influenced by environmental changes and variations in individual agent behaviour. This variation, while potentially advantageous in some contexts, may pose challenges in maintaining consistent performance in certain real-world applications.

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A Methodology

A.1 Algorithm Design

Algorithm 1: CIR3

Input : Max inner-refinement cycles L, Max outer-refinement cycles K, Max perspective M, Context c

Output : QA pairs $\mathcal{G}*$

$\mathcal{M} \leftarrow [\text{""}]$ `// Writer's short memory state.`

$\mathcal{H} \leftarrow [\text{""}]$ `// Long short-term memory state.`

`// Identify and assign unique perspectives` $\mathcal{P}$.

W_{P_0} = "default in-domain writer"

$W \leftarrow [W_{P_0}]$ `// List of Writers.`

`// List of subtopics` $\leq M$.

$P \leftarrow$ `classify_subtopics`(c, M)

foreach *subtopic in* P **do**

  W.append(`get_perspective_writer`(subtopic))

end

`// Outer-refinement cycles: evaluate and critic`

$k \leftarrow 0$

$\mathcal{F}\prime_{k+1} \leftarrow \varnothing$

do

  `// Inner-refinement cycles: generate, then refine.`

  $l \leftarrow 0$

  $\mathcal{F}_{l+1} \leftarrow \varnothing$

  do

    ${\mathcal{G}^+}_l \leftarrow \text{generateQAs}(c, \mathcal{M}[-1], \mathcal{H}[-1])$

    $\mathcal{F}_{l+1} \leftarrow \text{refineQAs}(({\mathcal{G}^+}_l, \mathcal{M}))$

    $\mathcal{M}.\text{append}(({\mathcal{G}^+}_l, \mathcal{F}_{l+1}))$

    $l++$

  while $l < L \wedge \mathcal{F}_{l+1} \neq \varnothing$;

  ${\mathcal{G}^-}_k \leftarrow {\mathcal{G}^+}_{l-1}$

  $\mathcal{F}\prime_{k+1} \leftarrow \text{curmudgeonQAs}(({\mathcal{G}^-}_k, \mathcal{H}))$

  $\mathcal{H}.\text{append}(({\mathcal{G}^-}_k, \mathcal{F}\prime_{k+1}))$

  $k++$

while $k < K \wedge \mathcal{F}\prime_{k+1} \neq \varnothing$;

$\mathcal{G}^* \leftarrow {\mathcal{G}^-}_{k-1}$

return $\mathcal{G}^*$

A.2 Collective Cognition

Collective Cognitive Convergence (C3) (Parunak et al., 2008, 2009) is a sociology and evolutionary-biology related phenomenon that occurs when frequent interactions among a group lead to a convergence in cognitive[6] orientation. This phenomenon is observed in various contexts, including research sub-disciplines, political and religious associations, and persistent adversarial configurations. The authors highlight that while convergence facilitates mutual understanding and coordination, if left unchecked, it can lead to cognitive collapse, by reducing the diversity of concepts to which the group is exposed. Hence, limiting the group's ability to explore other viewpoints and generate new ideas. The authors suggest that the lack of mechanism to introduce variation incites the collapse and reduction in diversity. To offset the natural tendency to converge, C3 explored various mechanisms for adding variation, from which the curmudgeon[7] modelling demonstrates a robust and effective mechanism of keeping the group from collapsing while converging.

(Woolley et al., 2015; Woolley and Gupta, 2024) investigates how group characteristics, including composition and structure, explain variations in their Collective Intelligence (CI). The authors highlight that the concept of CI in groups emerges from the interplay of *bottom-up* and *top-down* processes. Bottom-up processes involve the aggregation of individual member characteristics, such as intelligence, social skills, and diverse knowledge, which contribute to enhanced group collaboration. Top-down processes include structures, norms, and routines that regulate collective behaviour and can either enhance or impede coordination and collaboration within the group. The interaction and combination of these bottom-up and top-down aspects ultimately determine the level of a group's collective intelligence and performance.

B Corpus Details

As highlighted in 4.1, existing QA datasets lack comprehensive sets of QA pairs for input doc-

[6] Teasley et al. (2008) *uses the term "cognitive convergence" to encompass various concepts that have been used to explain the important processes underlying successful collaboration, such as inter-subjectivity, co-construction, knowledge convergence, common ground, joint problem space, and transactive reasoning.*

[7] As defined by (Parunak et al., 2009) *A curmudgeon is a non-conformist, someone who regularly questions the group's norms and assumptions.*

uments, making them unsuitable for evaluating CIR3. While MONSERRATE dataset (Rodrigues et al., 2022) does offer a set of questions for each short sentence context, it has several limitations: (1) it does not include answers, (2) its contexts are limited to short sentences, (3) questions are often paraphrases rather than semantically distinct, and (4) some questions are unanswerable based on the given context. Therefore, we opt to experiment solely with passages from the following datasets:

1. **FiQA** (Maia et al., 2018) dataset[8] was employed in the Financial Opinion Mining and Question Answering challenge at the 2018 International World Wide Web Conference. FiQA comprises 6,648 questions and 57,640 answer passages. It was curated from financial posts on platforms like Stackexchange[9], Reddit[10], and StockTwits[11] between 2009 and 2017, with the objective of developing question-answering systems that can address financial queries by leveraging information from diverse sources such as microblogs, reports, and news articles.

2. **InsuranceQA** (Feng et al., 2015) corpus[12], sourced from the Insurance Library[13] website, consists of 16,889 real-world user questions and 27,413 corresponding answers written by professionals with extensive domain knowledge in the insurance industry.

C Automatic Evaluation

C.1 Automatic Metrics

We provide a brief description of the metrics used in this study:

- **ROUGE-L** (Lin, 2004) assesses recall by evaluating the overlap between reference and generated sentences using Longest Common Subsequence statistics. We use the implementation from GOOGLE[14]. In this paper, we report the F1 score, the harmonic mean of precision and recall.

- **METEOR** (Banerjee and Lavie, 2005) is a recall-oriented metric that measures the similarity between generated and reference text, incorporating synonyms, stemming, and paraphrasing. We use the implementation from NLTK[15].

- **Jaccard Index** or **Jaccard coefficient** is a measure of similarity between two sets. It is calculated as the size of their intersection (elements they share) divided by the size of their union (total unique elements). Values range from 0 (no similarity) to 1 (identical sets). We adopt SCIKIT-LEARN's implementation[16].

- **BERTScore**[17] (Zhang et al., 2020) utilises BERT's contextual embeddings to assess word-level similarity via cosine similarity, correlating with human judgement in sentence and system evaluation, and providing precision, recall, and F1 metrics.

- **Vendi Score**[18] (Friedman and Dieng, 2023) is a similarity-based diversity metric derived from quantum statistical mechanics. It quantifies diversity while accounting for item similarity and not requiring prevalence information.

C.2 Score Calculations

We denote $s(c, \mathcal{Q})$, $s(c, \mathcal{A})$, $s(\mathcal{Q}, \mathcal{A})$, $s(c, \mathcal{Q}_{\oplus})$, $s(c, \mathcal{A}_{\oplus})$, and $s(\mathcal{Q}_{\oplus}, \mathcal{A}_{\oplus})$ the scores between (context and questions), (context and answers), (questions and answers), (context and concatenated questions), (context and concatenated answers), and (concatenated questions and concatenated answers), respectively. The scores are calculated as follow:

[8] https://huggingface.co/datasets/BeIR/fiqa
[9] https://stackexchange.com/
[10] https://www.reddit.com/
[11] https://stocktwits.com/
[12] https://github.com/shuzi/insuranceQA
[13] https://www.insurancelibrary.com/
[14] https://pypi.org/project/rouge-score
[15] https://www.nltk.org
[16] https://scikit-learn.org
[17] https://github.com/Tiiiger/bert_score
[18] https://github.com/vertaix/Vendi-Score

$$s(c, \mathcal{Q}) = \frac{1}{N}\sum_{i=1}^{N} s(c, q_i) \quad (1)$$

$$s(c, \mathcal{A}) = \frac{1}{N}\sum_{i=1}^{N} s(c, a_i) \quad (2)$$

$$s(\mathcal{Q}, \mathcal{A}) = \frac{1}{N}\sum_{i=1}^{N} s(q_i, a_i) \quad (3)$$

$$s(c, \mathcal{Q}_{\oplus}) = s(c, \oplus_{i=1}^{N} q_i) \quad (4)$$

$$s(c, \mathcal{A}_{\oplus}) = s(c, \oplus_{i=1}^{N} a_i) \quad (5)$$

$$s(\mathcal{Q}_{\oplus}, \mathcal{A}_{\oplus}) = s(\oplus_{i=1}^{N} q_i, \oplus_{i=1}^{N} a_i) \quad (6)$$

where s is the scoring function, and $\oplus$ is the concatenation function.

C.3 LLM-based Scorers

Recent research has explored using LLMs as evaluators (Kocmi and Federmann, 2023; Fu et al., 2023; Wang et al., 2023; Kim et al., 2024). This approach involves LLMs scoring the candidate output based on its generation probability.

G-EVAL[19] (Liu et al., 2023), a promising LLM-based evaluator, has shown significantly better correlation with human evaluations compared to existing methods. It consists of three key components: (1) a prompt defining the task and evaluation criteria, (2) Chain-of-Thoughts (CoT) (Wei et al., 2023) generating detailed evaluation steps, and (3) a scoring function that calls LLM, then calculates the score based on the probabilities of the returned tokens.

Figure 7 and 8 show the prompts used in our LLM-based evaluation.

D Human Evaluation

To further evaluate the quality of the question-answer pairs, we conducted human evaluation study. We recruited 8 annotators (volunteers) with 2 to 6 years of experience in the finance domain, all based in Europe. Each annotator was tasked with assessing 10 sets of question-answer pairs, focusing on comprehensiveness and faithfulness. Comprehensiveness was evaluated based on three key aspects: *coverage*, *depth*, and *coherence*, while faithfulness was assessed based on: *accuracy*, *representation*, and *diversification*.

For each aspect, the annotators assigned a score on a scale from 1 (worst) to 5 (best), based on the following guidelines:

> This document describes the instructions of human evaluation for AI-based question-answer generation task. The results will be made publicly available within 12 months of study completion.
>
> By proceeding with this task, you agree that any resulting work product may be shared publicly.
>
> You will be provided with two datasets, each containing a collection of text documents and corresponding question-answer pairs. Your task is to evaluate the question-answer pairs in terms of their accuracy and completeness in relation to the information presented in the associated text documents.
>
> **Instructions:**
>
> **Familiarise yourself with the context:** Carefully read the provided context to understand the topic and key information it contains.
>
> **Review the generated question-answer pairs:** Examine each question-answer pair.
>
> **Evaluate each criterion:** For each of the following criteria, rate the question-answer pair on a scale of **1** to **5** (5 being the highest), and provide a brief explanation for your rating:
>
> - **Coverage**: Does the question-answer pair address the main ideas and important details in the context?
> - **Depth:** Does the question prompt deeper understanding of the context, or is it superficial? Does the answer provide sufficient detail and explanation?
> - **Accuracy:** Is the answer factually correct and complete based on the information in the context?
> - **Coherence:** Does the question-answer pair flow logically? Does the question naturally lead to the answer, and do they together contribute to a better understanding of the topic?
> - **Representation:** Does the question-answer pair distort or present misleadingly any facts in the context?
> - **Diversification:** Does the collection of question-answer pairs provide diverse and unique insights, or is there significant overlap in the knowledge they convey?

Note: This section represents a partial excerpt of the complete evaluation guidelines.

E Prompts

Figure 3, 4, 5, and 6 show the illustrative prompts used in our implementation.

We also release CIR3's source code on GitHub[20].

[19] https://github.com/nlpyang/geval

[20] https://github.com/anonym-nlp-ai/cirrr

LLM Prompt

Task: Analyse the input document and identify its main topic and up to {M} subtopics it covers.

Input:

- document: {document}

Output:
Format the output as JSON with the following key:

- subtopics: [list of identified subtopics]

Example demonstrations:

document: At first glance, life insurance and annuities might seem like opposites. Life insurance is primarily used to pay your heirs when you pass away. An annuity grows your savings and pays you income while you're still alive. However, some life insurance policies let you build savings while alive, and annuities can include a death benefit payment.
subtopics: [finance, life insurance, annuity, saving]

document: Homeowners insurance and mortgage insurance are both types of insurance that can add to the cost of owning property, and you're likely to encounter both during the mortgage process. However, that's where their similarity ends. Homeowners insurance protects your home and its contents, while mortgage insurance, also called private mortgage insurance (PMI), protects your mortgage lender in case you default on your mortgage payments.[a]
subtopics: [finance, homeowners insurance, mortgage insurance, mortgage]

[a] Demonstration passages are sourced from https://www.investopedia.com.

Figure 3: General illustration of the *classifier* agent prompt in CIR3, corresponding to Line 7 in Algorithm 1.

LLM Prompt

Task: You have been provided with a list of subtopics and a document. Your role is to manage a group of {M} writer agents to generate a comprehensive and faithful set of question-answer pairs given the input document.

Instructions:

1. Assign each subtopic as a unique perspective to a different writer agent.
2. Instruct the writers to analyse the document in-depth from their assigned perspective.
3. Each writer must generate up to {N} question-answer pairs that are directly grounded in the document's content.
4. After the initial generation, instruct the agents to reflect on their question-answer pairs with respect to the input document. If writers agree that no refinements are needed, they should return "agreement" as feedback, otherwise "continue".

Input:

- Subtopics: {subtopics}
- Document: {document}
- Number of question-answer pairs to be generated: {N}

Output:
Format the output as JSON with the following keys:

- instructions_list: [list of instruction to the writers]

Figure 4: General illustration of the *moderator* agent prompt in CIR3, corresponding to Line 8, 9, 10 and initial iteration in 16-23 in Algorithm 1.

LLM Prompt

Task: Refine the list of your generated question-answer pairs based on the group feedback.

Instructions:

1. Review and refine all generated question-answer pairs based on the provided feedback from other writer agents, while ensuring that the question-answer pairs:
    - Accurately represent your assigned subtopic.
    - Are well-supported by the document's content.
    - Offer diverse and comprehensive coverage of the subtopic within the context of the document.
2. If you agree that no refinements are needed, return "agreement" as feedback, otherwise "continue".
3. Output the refined list of question-answer pairs along with your feedback.

Input:

- Document: {document}
- question_answer_list: {question_answer_list}
- group_feedback: {feedback_list}

Output:
Format the output as JSON with the following keys:

- question_answer_list: [list of refined question-answer pairs]
- group_feedback: [list of new feedback]
- status: continue or agreement.

Figure 5: General illustration of the *writer* agents prompt in CIR3, corresponding to Line 18-23 in Algorithm 1.

LLM Prompt

Task: Your role is to assess the quality of the generated question-answer pairs generated by the writer agents.

Instructions:

- Review the generated question-answer pairs in relation to the original document and your previous feedback.
- Ensure the pairs are comprehensive, diverse, grounded in the document, and free from assumptions.
- For diversity, use your judgement to determine whether to provide feedback based on your own knowledge, scores from an external evaluation tool, or both. The evaluation tool scores quantify the diversity of (a) generated questions (score_1), (b) generated answers (score_2), and (c) concatenated answers and input document (score_3). The objective of your feedback on diversity is to minimise score_3, while maximising score_1 and score_3.
- Provide constructive feedback on each pair, highlighting any areas for improvement.
- If you are satisfied with the quality, return "agreement".
- If further refinement is needed, instruct the writers to begin another round of generation and refinement by returning the feedback and the status "continue".

Input:

- Document: {document}
- question_answer_list: {list of question answers}
- old_feedback: {list of previous feedback lists}

Output:
Format the output as JSON with the following keys:

- feedback: [list of new feedback]
- status: continue or agreement.

Figure 6: General illustration of the *curmudgeon* agent prompt in CIR3, corresponding to Line 11-13 and 14-29 in Algorithm 1.

LLM Prompt

Analyse the comprehensiveness of the generated QA pairs in relation to the given context. Follow this structured approach:

1. **Coverage:** Assess if the QA pairs address the main ideas and important details in the context.
2. **Depth:** Evaluate whether the questions cover relevant aspects of the context and prompt deeper understanding.
3. **Accuracy:** Determine the correctness and completeness of the answers based on the information in the context.
4. **Coherence:** Assess the overall flow and logical connection between the questions and answers.

Provide a score for each criterion (1-5, with 5 being the highest), along with a thorough explanation of your reasoning. Conclude with an overall assessment of the QA pair's comprehensiveness.

Context: {Context}

Generated QA Pairs:

1. Q: {Question 1}
   A: {answer 1}
2. Q: {Question 2}
   A: {answer 2}

i. Q: {Question i}
   A: {Answer i}

N. Q: {{Question N}}
   A: {answer N}

Figure 7: Illustrative prompt designed for evaluating CIR3 comprehensiveness.

LLM Prompt

Analyse the faithfulness and potential for hallucination in the generated QA pairs in relation to the given context. Follow this structured approach:

1. **Accuracy:** Assess whether the questions and answers accurately reflect the information provided in the context.
2. **Exaggeration:** Determine if any claims in the answers overstate or embellish the information from the context.
3. **Consistency:** Determine if the answers contradict or deviate from the facts presented in the context.
4. **Justification:** Evaluate whether the questions and answers are well-supported by evidence from the context.
5. **Plausibility:** Assess whether the questions and answers, even if not explicitly stated, are reasonable inferences based on the context.
6. **Misrepresentation:** Assess if any facts in the context are distorted or presented misleadingly in the answers.

Provide a score for each criterion (1-5, with 5 being the highest), along with a thorough explanation of your reasoning, including specific examples from the context and generated QA pairs. Conclude with an overall assessment of the QA pair's faithfulness and potential for hallucination.

Context: {Context}

Generated QA Pairs:

1. Q: {Question 1}
   A: {answer 1}
2. Q: {Question 2}
   A: {answer 2}

i. Q: {Question i}
   A: {Answer i}

N. Q: {{Question N}}
   A: {answer N}

Figure 8: Illustrative prompt designed for evaluating CIR3 faithfulness and hallucination.